

Orthogonal Projections of Functions on Polynomials: A Method for Integral Inequalities

Theoretical Framework and Intuition

In many integral inequality problems, we aim to find the minimum value of $\int_a^b (f(x))^2 dx$ given constraints of the form $\int_a^b f(x)x^k dx = c_k$. The core idea is to **approximate the unknown function $f(x)$ by a polynomial $P(x)$ of a suitable degree** that 'imitates' the behavior of $f(x)$ regarding these exact integral conditions.

After that, we employ the **Cauchy-Schwarz Inequality for Integrals**, which says that for square-integrable functions f, g on $[a, b]$,

$$\left(\int_a^b (f(x))^2 dx \right) \left(\int_a^b (g(x))^2 dx \right) \geq \left(\int_a^b f(x)g(x) dx \right)^2.$$

Equality holds if and only if $f(x) = \lambda g(x)$ almost everywhere.

The Method

1. **Choose the Degree:** If given conditions on $\int f(x) dx, \dots, \int x^k f(x) dx$, choose a degree k polynomial

$$P(x) = a_0 + a_1x + \dots + a_kx^k$$

2. **Imitate the Function:** Force the polynomial to satisfy the exact same conditions as $f(x)$:

$$\int_a^b P(x)x^j dx = \int_a^b f(x)x^j dx \quad \text{for } j = 0, 1, \dots, k$$

This generates a system of linear equations to solve for the coefficients $a_0, a_1, \dots, a_k$.

3. **Apply Cauchy-Schwarz:** Apply the inequality on $f(x)$ and $P(x)$:

$$\left(\int_a^b (f(x))^2 dx \right) \left(\int_a^b (P(x))^2 dx \right) \geq \left(\int_a^b f(x)P(x) dx \right)^2$$

Since $P(x)$ is a linear combination of $1, x, \dots, x^k$, the cross-term $\int_a^b f(x)P(x) dx$ is completely determined by the initial conditions and equals $\int_a^b (P(x))^2 dx$. This yields a sharp lower bound.

Demonstrative Examples

Example 1

Let $f : [0, 1] \rightarrow \mathbb{R}$ be a continuous function such that $\int_0^1 f(x) dx = 1$. Find the minimum value of $\int_0^1 (f(x))^2 dx$.

Solution. We approximate $f(x)$ by a degree 0 polynomial: $P(x) = c$. Imposing the condition:

$$\begin{aligned} \int_0^1 c dx &= 1 \\ \implies c &= 1 \end{aligned}$$

Thus, $P(x) = 1$. Applying Cauchy-Schwarz to $f(x)$ and $P(x)$:

$$\begin{aligned} \left(\int_0^1 (f(x))^2 dx \right) \left(\int_0^1 1^2 dx \right) &\geq \left(\int_0^1 f(x) \cdot 1 dx \right)^2 \\ \implies \left(\int_0^1 (f(x))^2 dx \right) (1) &\geq (1)^2 \\ \implies \int_0^1 (f(x))^2 dx &\geq 1 \end{aligned}$$

Equality holds for $f(x) = 1$. □

Example 2

Let $f : [0, 1] \rightarrow \mathbb{R}$ be continuous, $\int_0^1 f(x) dx = 1$, and $\int_0^1 x f(x) dx = 1$. Find the minimum value of $\int_0^1 (f(x))^2 dx$.

Solution. We approximate f with $P(x) = ax + b$. Imposing the conditions:

$$\begin{aligned} \int_0^1 (ax + b) dx &= 1 \\ \implies \frac{a}{2} + b &= 1 \end{aligned}$$

$$\begin{aligned} \int_0^1 x(ax + b) dx &= 1 \\ \implies \frac{a}{3} + \frac{b}{2} &= 1 \end{aligned}$$

Solving this system by expressing b in terms of a :

$$\begin{aligned}
 b &= 1 - \frac{a}{2} \\
 \Rightarrow \frac{a}{3} + \frac{1}{2} \left(1 - \frac{a}{2}\right) &= 1 \\
 \Rightarrow \frac{a}{3} + \frac{1}{2} - \frac{a}{4} &= 1 \\
 \Rightarrow \frac{a}{12} &= \frac{1}{2} \\
 \Rightarrow a &= 6
 \end{aligned}$$

Substituting back gives $b = -2$. So, $P(x) = 6x - 2$. We compute the integral of its square:

$$\begin{aligned}
 \int_0^1 (6x - 2)^2 dx &= \left[\frac{(6x - 2)^3}{18} \right]_0^1 \\
 \Rightarrow \int_0^1 (6x - 2)^2 dx &= \frac{64}{18} - \left(-\frac{8}{18} \right) \\
 \Rightarrow \int_0^1 (6x - 2)^2 dx &= \frac{72}{18} = 4
 \end{aligned}$$

Now, apply Cauchy-Schwarz on $f(x)$ and $P(x)$:

$$\left(\int_0^1 (f(x))^2 dx \right) \left(\int_0^1 (6x - 2)^2 dx \right) \geq \left(\int_0^1 f(x)(6x - 2) dx \right)^2$$

Evaluating the cross-term using the given conditions:

$$\begin{aligned}
 \int_0^1 f(x)(6x - 2) dx &= 6 \int_0^1 xf(x) dx - 2 \int_0^1 f(x) dx \\
 \Rightarrow \int_0^1 f(x)(6x - 2) dx &= 6(1) - 2(1) \\
 \Rightarrow \int_0^1 f(x)(6x - 2) dx &= 4
 \end{aligned}$$

Thus, the inequality simplifies to:

$$\begin{aligned}
 \left(\int_0^1 (f(x))^2 dx \right) (4) &\geq (4)^2 \\
 \Rightarrow \int_0^1 (f(x))^2 dx &\geq 4
 \end{aligned}$$

□

Example 3

Let $f : [-1, 1] \rightarrow \mathbb{R}$ be continuous, $\int_{-1}^1 f(x) dx = 0$, and $\int_{-1}^1 xf(x) dx = 1$. Find the minimum of $\int_{-1}^1 (f(x))^2 dx$.

Solution. We choose $P(x) = ax + b$. Imposing the conditions:

$$\begin{aligned}\int_{-1}^1 (ax + b) dx &= 0 \\ \Rightarrow \left[\frac{ax^2}{2} + bx \right]_{-1}^1 &= 0 \\ \Rightarrow 2b &= 0 \\ \Rightarrow b &= 0\end{aligned}$$

$$\begin{aligned}\int_{-1}^1 x(ax) dx &= 1 \\ \Rightarrow a \left[\frac{x^3}{3} \right]_{-1}^1 &= 1 \\ \Rightarrow \frac{2a}{3} &= 1 \\ \Rightarrow a &= \frac{3}{2}\end{aligned}$$

Thus, $P(x) = \frac{3}{2}x$. We compute the integral of its square:

$$\begin{aligned}\int_{-1}^1 (P(x))^2 dx &= \int_{-1}^1 \frac{9}{4}x^2 dx \\ \Rightarrow \int_{-1}^1 (P(x))^2 dx &= \frac{9}{4} \left[\frac{x^3}{3} \right]_{-1}^1 \\ \Rightarrow \int_{-1}^1 (P(x))^2 dx &= \frac{9}{4} \cdot \frac{2}{3} = \frac{3}{2}\end{aligned}$$

By Cauchy-Schwarz:

$$\begin{aligned}\left(\int_{-1}^1 (f(x))^2 dx \right) \left(\int_{-1}^1 \left(\frac{3}{2}x \right)^2 dx \right) &\geq \left(\int_{-1}^1 \frac{3}{2}xf(x) dx \right)^2 \\ \Rightarrow \left(\int_{-1}^1 (f(x))^2 dx \right) \left(\frac{3}{2} \right) &\geq \left(\frac{3}{2} \cdot 1 \right)^2 \\ \Rightarrow \int_{-1}^1 (f(x))^2 dx &\geq \frac{3}{2}\end{aligned}$$

□

Example 4

Let f be continuously differentiable on $[0, 1]$ such that $f(0) = 0$, $f(1) = 1$, and $\int_0^1 f(x) dx = 1$. Minimize $\int_0^1 (f'(x))^2 dx$.

Solution. We must translate the conditions on f into conditions on f' . By the Fundamental Theorem of Calculus:

$$\begin{aligned}\int_0^1 f'(x) dx &= f(1) - f(0) \\ \implies \int_0^1 f'(x) dx &= 1\end{aligned}$$

Using Integration by Parts on the integral condition:

$$\begin{aligned}\int_0^1 f(x) dx &= 1 \\ \implies [xf(x)]_0^1 - \int_0^1 xf'(x) dx &= 1 \\ \implies (1 \cdot 1 - 0) - \int_0^1 xf'(x) dx &= 1 \\ \implies 1 - \int_0^1 xf'(x) dx &= 1 \\ \implies \int_0^1 xf'(x) dx &= 0\end{aligned}$$

We project f' onto $P(x) = ax + b$:

$$\begin{aligned}\int_0^1 (ax + b) dx &= 1 \implies \frac{a}{2} + b = 1 \\ \int_0^1 x(ax + b) dx &= 0 \implies \frac{a}{3} + \frac{b}{2} = 0 \implies b = -\frac{2a}{3}\end{aligned}$$

Substituting b into the first equation:

$$\begin{aligned}\frac{a}{2} - \frac{2a}{3} &= 1 \\ \implies -\frac{a}{6} &= 1 \\ \implies a &= -6\end{aligned}$$

This gives $b = 4$, so $P(x) = -6x + 4$. Now, we square the polynomial:

$$\begin{aligned}\int_0^1 P(x)^2 dx &= \int_0^1 (-6x + 4)^2 dx \\ \implies \int_0^1 P(x)^2 dx &= \int_0^1 (36x^2 - 48x + 16) dx \\ \implies \int_0^1 P(x)^2 dx &= 12 - 24 + 16 = 4\end{aligned}$$

Applying Cauchy-Schwarz to $f'(x)$ and $P(x)$ yields:

$$\begin{aligned} \left(\int_0^1 (f'(x))^2 dx \right) (4) &\geq (4)^2 \\ \implies \int_0^1 (f'(x))^2 dx &\geq 4 \end{aligned}$$

□

Example 5

Let $f : [0, 1] \rightarrow \mathbb{R}$ be continuous, $\int_0^1 f(x) dx = 1$, and $\int_0^1 x^2 f(x) dx = \frac{1}{2}$. Minimize $\int_0^1 (f(x))^2 dx$.

Solution. We project onto $P(x) = a + bx^2$:

$$\begin{aligned} \int_0^1 (a + bx^2) dx = 1 &\implies a + \frac{b}{3} = 1 \\ \int_0^1 x^2(a + bx^2) dx = \frac{1}{2} &\implies \frac{a}{3} + \frac{b}{5} = \frac{1}{2} \end{aligned}$$

Multiplying the first equation by $\frac{1}{3}$:

$$\frac{a}{3} + \frac{b}{9} = \frac{1}{3}$$

Subtracting this from the second equation:

$$\begin{aligned} b \left(\frac{1}{5} - \frac{1}{9} \right) &= \frac{1}{2} - \frac{1}{3} \\ \implies b \left(\frac{4}{45} \right) &= \frac{1}{6} \\ \implies b &= \frac{1}{6} \cdot \frac{45}{4} = \frac{15}{8} \end{aligned}$$

Substituting b back to find a :

$$\begin{aligned} a &= 1 - \frac{1}{3} \left(\frac{15}{8} \right) \\ \implies a &= 1 - \frac{5}{8} = \frac{3}{8} \end{aligned}$$

Thus, $P(x) = \frac{3}{8} + \frac{15}{8}x^2$. The cross-term evaluates to:

$$\begin{aligned}\int_0^1 P(x)f(x) dx &= \int_0^1 \left(\frac{3}{8} + \frac{15}{8}x^2 \right) f(x) dx \\ \implies \int_0^1 P(x)f(x) dx &= \frac{3}{8} \int_0^1 f(x) dx + \frac{15}{8} \int_0^1 x^2 f(x) dx \\ \implies \int_0^1 P(x)f(x) dx &= \frac{3}{8}(1) + \frac{15}{8} \left(\frac{1}{2} \right) \\ \implies \int_0^1 P(x)f(x) dx &= \frac{6}{16} + \frac{15}{16} = \frac{21}{16}\end{aligned}$$

By Cauchy-Schwarz on $f(x)$ and $P(x)$:

$$\begin{aligned}\left(\int_0^1 (f(x))^2 dx \right) \left(\frac{21}{16} \right) &\geq \left(\frac{21}{16} \right)^2 \\ \implies \int_0^1 (f(x))^2 dx &\geq \frac{21}{16}\end{aligned}$$

□

Practice Problems

1. Let $f : [-1, 1] \rightarrow \mathbb{R}$ be a continuous function such that $\int_{-1}^1 f(x) dx = 2$ and $\int_{-1}^1 xf(x) dx = 0$.
 2. Find the minimum possible value of $\int_{-1}^1 (f(x))^2 dx$ and determine the function $f(x)$ that achieves this minimum.
2. Let $f : [0, 1] \rightarrow \mathbb{R}$ be a continuously differentiable function such that $f(0) = f(1) = 0$ and $\int_0^1 f(x) dx = 1$. Prove that $\int_0^1 (f'(x))^2 dx \geq 12$.
3. Let $f : [0, 1] \rightarrow \mathbb{R}$ be a continuous function satisfying $\int_0^1 xf(x) dx = 1$ and $\int_0^1 x^2 f(x) dx = 1$.
 1. Establish a sharp lower bound for $\int_0^1 (f(x))^2 dx$.
4. Let $f : [0, \pi] \rightarrow \mathbb{R}$ be a continuous function such that $\int_0^\pi f(x) \sin x dx = 1$ and $\int_0^\pi f(x) \cos x dx = 1$. Find the minimum value of $\int_0^\pi (f(x))^2 dx$.
5. Let f be a continuously differentiable function on $[0, 1]$ such that $f(0) = 0$ and $f(1) = 1$.
 1. Evaluate the minimum value of $\int_0^1 (f'(x))^2 dx$ given the additional constraint that $\int_0^1 xf(x) dx = \frac{1}{3}$.
6. Let $f : [0, 1] \rightarrow \mathbb{R}$ be continuous. Given that $\int_0^1 f(x) dx = a$ and $\int_0^1 xf(x) dx = b$ for some real constants a and b , prove that:

$$\int_0^1 (f(x))^2 dx \geq 4a^2 - 12ab + 12b^2.$$

7. Let f be a twice continuously differentiable function on $[0, 1]$ such that $f(0) = f'(0) = 0$ and $f(1) = 1$. Prove that $\int_0^1 (f''(x))^2 dx \geq 4$.